

Paul's Online Notes

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Section 4.10 : Table Of Laplace Transforms

Table of Laplace Transforms

$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1. 1	$\frac{1}{s}$
2. e^{at}	$\frac{1}{s-a}$
3. $t^n, n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
4. $t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5. $\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$
6. $t^{n-\frac{1}{2}}, n = 1, 2, 3, \dots$	$\frac{1 \cdot 3 \cdot 5 \cdots (2n-1) \sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7. $\sin(at)$	$\frac{a}{s^2 + a^2}$
8. $\cos(at)$	$\frac{s}{s^2 + a^2}$
9. $t \sin(at)$	$\frac{2as}{(s^2 + a^2)^2}$
10. $t \cos(at)$	$\frac{s^2 - a^2}{(s^2 + a^2)^2}$
11. $\sin(at) - at \cos(at)$	$\frac{2a^3}{(s^2 + a^2)^2}$
12. $\sin(at) + at \cos(at)$	$\frac{2as^2}{(s^2 + a^2)^2}$
13. $\cos(at) - at \sin(at)$	$\frac{s(s^2 - a^2)}{(s^2 + a^2)^2}$

14. $\cos(at) + at \sin(at)$	$\frac{s(s^2 + 3a^2)}{(s^2 + a^2)^2}$
15. $\sin(at + b)$	$\frac{s \sin(b) + a \cos(b)}{s^2 + a^2}$
16. $\cos(at + b)$	$\frac{s \cos(b) - a \sin(b)}{s^2 + a^2}$
17. $\sinh(at)$	$\frac{a}{s^2 - a^2}$
18. $\cosh(at)$	$\frac{s}{s^2 - a^2}$
19. $e^{at} \sin(bt)$	$\frac{b}{(s - a)^2 + b^2}$
20. $e^{at} \cos(bt)$	$\frac{s - a}{(s - a)^2 + b^2}$
21. $e^{at} \sinh(bt)$	$\frac{b}{(s - a)^2 - b^2}$
22. $e^{at} \cosh(bt)$	$\frac{s - a}{(s - a)^2 - b^2}$
23. $t^n e^{at}, \quad n = 1, 2, 3, \dots$	$\frac{n!}{(s - a)^{n+1}}$
24. $f(ct)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
25. $u_c(t) = u(t - c)$ Heaviside Function	$\frac{e^{-cs}}{s}$
26. $\delta(t - c)$ Dirac Delta Function	e^{-cs}
27. $u_c(t) f(t - c)$	$e^{-cs} F(s)$
28. $u_c(t) g(t)$	$e^{-cs} \mathcal{L}\{g(t + c)\}$
29. $e^{ct} f(t)$	$F(s - c)$
30. $t^n f(t), \quad n = 1, 2, 3, \dots$	$(-1)^n F^{(n)}(s)$
31. $\frac{1}{t} f(t)$	$\int_s^\infty F(u) du$
32. $\int_0^t f(v) dv$	$\frac{F(s)}{s}$

$$33. \int_0^t f(t-\tau) g(\tau) d\tau$$

$$F(s) G(s)$$

$$34. f(t+T) = f(t)$$

$$\frac{\int_0^T e^{-st} f(t) dt}{1 - e^{-sT}}$$

$$35. f'(t)$$

$$sF(s) - f(0)$$

$$36. f''(t)$$

$$s^2 F(s) - sf(0) - f'(0)$$

$$37. f^{(n)}(t)$$

$$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

Table Notes

1. This list is not a complete listing of Laplace transforms and only contains some of the more commonly used Laplace transforms and formulas.
2. Recall the definition of hyperbolic functions.

$$\cosh(t) = \frac{e^t + e^{-t}}{2} \quad \sinh(t) = \frac{e^t - e^{-t}}{2}$$

3. Be careful when using “normal” trig function vs. hyperbolic functions. The only difference in the formulas is the “ $+a^2$ ” for the “normal” trig functions becomes a “ $-a^2$ ” for the hyperbolic functions!
4. Formula #4 uses the Gamma function which is defined as

$$\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} dx$$

If n is a positive integer then,

$$\Gamma(n+1) = n!$$

The Gamma function is an extension of the normal factorial function. Here are a couple of quick facts for the Gamma function

$$\Gamma(p+1) = p\Gamma(p)$$

$$p(p+1)(p+2)\cdots(p+n-1) = \frac{\Gamma(p+n)}{\Gamma(p)}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$